

Minimum Error Rate Classification

Bayesian Decision Theory — Clean rewrite with explanations, worked examples, and the Minimax criterion

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1. What "Minimum Error Rate" Means

A classifier makes a **misclassification error** whenever it picks the wrong class for a data point. "Minimum error rate classification" is simply the classification strategy that **minimizes the probability of making this kind of mistake**, averaged over all data points.

Setting up the idea

Suppose we have patients and a model that predicts whether each one is **Healthy** or has a **Disease**:

Patient	Actual Class
P1	Healthy
P2	Healthy
P3	Disease
P4	Disease

Now a **new patient** arrives. The model computes **posterior probabilities** from the observed data x :

$$P(\text{Healthy} \mid x) = 0.30 \quad P(\text{Disease} \mid x) = 0.70$$

Since $0.70 > 0.30$, the decision is: **classify as Disease**.

Why this works: If we always pick the class with the higher posterior probability, we are, on average, right more often than any other rule could make us. That is exactly what "minimum error rate" means.

Formal definition

Let $C_1 = \text{Healthy}$, $C_2 = \text{Disease}$.
A **classification error** happens when we choose the wrong class for a given input.

$$\text{Max Posterior Probability Rule} = \text{Minimum Error Rate Classification}$$

So: always choose the class with the largest posterior probability $P(C_i \mid x)$.

2. The Decision Rule & Likelihood Ratio Test

Basic decision rule

$$\text{If } P(C_1 \mid x) > P(C_2 \mid x) \Rightarrow \text{decide } x \in C_1$$

otherwise $\Rightarrow$ decide $x \in C_2$

This rule considers the classification error probability directly and picks whichever class minimizes it — it does **not** weigh different kinds of mistakes differently (that's what "cost" or "loss" would add, covered later under Minimax).

Likelihood ratio test

Using Bayes' theorem, $P(C_i | x) = p(x | C_i) P(C_i) / p(x)$. Since $p(x)$ is the same for both classes, comparing posteriors is the same as comparing $p(x | C_i) P(C_i)$. Define the likelihood ratio:

$$\Lambda(x) = p(x | C_1) / p(x | C_2)$$

If $\Lambda(x) > P(C_2)/P(C_1) \Rightarrow$ decide C_1

If $\Lambda(x) < P(C_2)/P(C_1) \Rightarrow$ decide C_2

This is just the same rule rearranged — it's useful because it separates "how the data looks under each class" (the likelihoods) from "how likely each class is beforehand" (the priors).

3. Worked Example 1 — Disease Diagnosis (Discrete Features)

Setup: C_1 = Diseased, C_2 = Healthy. Total = 1000 patients.

$$P(C_1) = 0.20 \quad P(C_2) = 0.80$$

Step 2 Feature: Temperature — counts of patients with High Temperature:

Class	Total	High Temp
C_1 (Diseased)	200	160
C_2 (Healthy)	800	120

So the likelihoods are: $P(\text{High} | C_1) = 160/200 = 0.80$, $P(\text{High} | C_2) = 120/800 = 0.15$

Step 3 Evidence — total probability of observing "High Temp" regardless of class:

$$\begin{aligned} P(\text{High}) &= P(\text{High}|C_1)P(C_1) + P(\text{High}|C_2)P(C_2) \\ &= (0.80 \times 0.20) + (0.15 \times 0.80) \\ &= 0.16 + 0.12 = \mathbf{0.28} \end{aligned}$$

Step 4 Posterior (Bayes' theorem):

$$P(C_1 | \text{High}) = (0.80 \times 0.20) / 0.28 = \mathbf{0.571}$$

$$P(C_2 | \text{High}) = (0.15 \times 0.80) / 0.28 = \mathbf{0.429}$$

Since $P(C_1|\text{High}) > P(C_2|\text{High})$, a patient with High Temperature is classified as **C_1 (Diseased)**.

Step 5 Error rate for this decision:

$$\text{error rate} = 1 - \max[P(C_1|x), P(C_2|x)] = 1 - 0.571 = \mathbf{0.429 = 42.9\%}$$

Interpretation: even though we pick the *most likely* class, we're still wrong 42.9% of the time for patients with high temperature — because the two classes overlap a lot on this feature. "Minimum error rate" doesn't mean zero error; it means the *smallest achievable* error given the overlap in the data.

4. Total Probability of Error & the Error Rate Formula

Zooming out from a single patient to the *whole feature space*: divide the space into decision regions. R_1 = region where we decide C_1 , R_2 = region where we decide C_2 . An error occurs in two ways:

- True class is C_1 , but x falls in R_2 (we decide C_2) — a **miss**
- True class is C_2 , but x falls in R_1 (we decide C_1) — a **false alarm**

$$P(\text{error}) = \int_{R_2} p(x|C_1) P(C_1) dx + \int_{R_1} p(x|C_2) P(C_2) dx$$

$R_1 \rightarrow$ region where the classifier chooses C_1

$R_2 \rightarrow$ region where the classifier chooses C_2

The minimum-error-rate rule works by choosing R_1 and R_2 so that this total integral is as small as possible — which turns out to be exactly the "pick whichever posterior is bigger" rule from Section 1.

5. Worked Example 2 — Continuous Gaussian Features (Pass/Fail)

Setup: C1 = Fail, C2 = Pass. x = study hours. Total = 1000 students.

$$P(C1) = 0.30 \quad P(C2) = 0.70$$

Study hours follow a **Normal (Gaussian) distribution** in each class:

$$p(x \mid C1) = N(\mu1 = 3, \sigma1 = 1) \quad p(x \mid C2) = N(\mu2 = 7, \sigma2 = 1)$$

Step 1 — Write the Gaussian likelihoods

$$p(x|C1) = (1/\sqrt{2\pi}) \exp(-(x-3)^2/2)$$
$$p(x|C2) = (1/\sqrt{2\pi}) \exp(-(x-7)^2/2)$$

Step 2 — Find the decision boundary

The boundary is the point where the two *posterior-weighted* curves cross:

$$p(x|C1)P(C1) = p(x|C2)P(C2)$$

Taking natural log on both sides (this turns the messy exponential equation into a simple quadratic):

$$\ln(0.30) - (x-3)^2/2 = \ln(0.70) - (x-7)^2/2$$

Using $\ln(0.30) = -1.204$ and $\ln(0.70) = -0.357$:

$$-1.204 - (x-3)^2/2 = -0.357 - (x-7)^2/2$$
$$(x-7)^2 - (x-3)^2 = 0.847 \times 2 = 1.694$$

Expanding both squares and simplifying (a difference of squares) gives a linear equation in x, which solves to:

$$x \approx 4.8 \text{ hours} \text{ — the decision boundary}$$

Step 3 — Define the decision regions

$x < 4.8 \Rightarrow$

decide C1 (Fail) — this is region R1

$x > 4.8 \Rightarrow$

decide C2 (Pass) — this is region R2

Step 4 — Compute the total error using the formula from Section 4

$$P(\text{error}) = \int_{R2} p(x|C1)P(C1) \, dx + \int_{R1} p(x|C2)P(C2) \, dx$$

Part 1 — true class is Fail (C1), but x falls in R2 (x > 4.8), so we wrongly say Pass:

$$= 0.30 \times \int_{4.8}^{\infty} (1/\sqrt{2\pi}) \exp(-(x-3)^2/2) \, dx$$

Convert to a **z-score** (standardize using $\mu=3, \sigma=1$):

$$z = (4.8 - 3)/1 = 1.8$$

$$\text{Part 1} = 0.30 \times P(Z > 1.8) = 0.30 \times 0.0359 = \mathbf{0.01077}$$

Part 2 — true class is Pass (C2), but x falls in R1 (x < 4.8), so we wrongly say Fail:

Here $z = (4.8 - 7)/1 = -2.2$, giving $P(Z < -2.2) = 0.00973$

Part 2 = **0.00973**

Final Error

$P(\text{error}) = \text{Part 1} + \text{Part 2} = 0.01077 + 0.00973 = 0.0205$

$P(\text{error}) = 2.05\%$

This is much lower than Example 1's 42.9% — because here the two classes ($\mu=3$ vs $\mu=7$, both $\sigma=1$) are far apart relative to their spread, so there's much less overlap and much less unavoidable confusion between them.

6. The Minimax Criterion

When do we need Minimax? The minimum-error-rate rule above assumes we know the prior probabilities $P(C1)$, $P(C2)$ reliably. But sometimes priors are **unknown or unreliable** (e.g. we don't actually know what fraction of visitors are intruders vs. authorized users). In that case, we can't optimize for the "average" case — instead we protect against the **worst case**.

Minimax = Minimize the Maximum possible error

Overall risk (Bayes Risk) with general costs

If wrong decisions have different costs λ_{ij} (cost of deciding C_i when the truth is C_j), the total expected risk is:

$$R = \int [\lambda_{11} P(C1) p(x|C1) + \lambda_{12} P(C2) p(x|C2)] dx \\ + \int [\lambda_{21} P(C1) p(x|C1) + \lambda_{22} P(C2) p(x|C2)] dx$$

Under the simple **0-1 loss** (correct = 0 cost, wrong = 1 cost, i.e. $\lambda_{11} = \lambda_{22} = 0$ and $\lambda_{12} = \lambda_{21} = 1$), this simplifies to:

$$R = P(C1) \cdot P(\text{error}|C1) + P(C2) \cdot P(\text{error}|C2)$$

$P(\text{error}|C1) = \int_{R2} p(x|C1) dx$ — the **miss rate**

$P(\text{error}|C2) = \int_{R1} p(x|C2) dx$ — the **false alarm rate**

The Minimax equation

$$R_{\text{minimax}} = \min_x \max_{P(C1)} R(P(C1))$$

Key condition: If $P(\text{error}|C1) = P(\text{error}|C2)$, then the boundary is "prior-proof" — it stays optimal no matter what the true prior turns out to be. This is the Minimax boundary: we deliberately choose the point where the miss rate equals the false alarm rate, so the *worst-case* error is as small as possible, regardless of which prior is actually correct.

7. Worked Example 3 — Minimax on a Security System

Setup: C1 = Intruder, C2 = Authorized Person. x = Voice Recognition Score (0 to 10).

Step 1 — Distributions

$$p(x|C1) = N(\mu_1 = 4, \sigma_1 = 1) \quad p(x|C2) = N(\mu_2 = 7, \sigma_2 = 1)$$

Step 2 — Prior probability is unknown

We genuinely don't know what fraction of people approaching the system are intruders vs. authorized — this can change over time and location. Since we can't rely on a fixed prior, we look for the threshold θ that keeps us safe no matter what the true prior is. That's exactly what Minimax gives us.

Step 3 — Conditional risks as a function of the threshold θ

$R1(\theta)$ = risk of wrongly classifying a true intruder (C1) as authorized = $P(Z > z1)$

$$z1 = (\theta - 4)/1 = \theta - 4$$

$R2(\theta)$ = risk of wrongly classifying a true authorized person (C2) as intruder = $P(Z < z2)$

$$z2 = (\theta - 7)/1 = \theta - 7$$

Step 4 — Table of R1 and R2 for different thresholds θ

θ	$z1 = \theta - 4$	$R1 = P(Z > z1)$	$z2 = \theta - 7$	$R2 = P(Z < z2)$
4.0	0.0	0.5000	-3.0	0.0013
4.5	0.5	0.3085	-2.5	0.0062
5.0	1.0	0.1587	-2.0	0.0228
5.5	1.5	0.0668	-1.5	0.0668
6.0	2.0	0.0228	-1.0	0.1587
6.5	2.5	0.0062	-0.5	0.3085
7.0	3.0	0.0013	0.0	0.5000

Step 5 — Find the Minimax point

The Minimax threshold is where $R1(\theta) = R2(\theta)$ — from the table, this happens at:

$$\theta_{\text{minimax}} = 5.5 \quad (R1 = R2 = 0.0668)$$

Step 6 — Minimax error

$$P(\text{error})_{\text{minimax}} = R1(\theta) = R2(\theta) = 0.0668 = \mathbf{6.68\%}$$

Step 7 — Overall Minimax risk (prior-free)

$$\begin{aligned} P_{\text{minimax}} &= P(C1) \times R1 + P(C2) \times R2 \\ &= P(C1) \times 0.0668 + P(C2) \times 0.0668 \\ &= 0.0668 \times [P(C1) + P(C2)] \\ &= 0.0668 \times 1 = \mathbf{0.0668 = 6.68\%} \end{aligned}$$

Notice the priors cancel out entirely — because we chose θ so that $R1 = R2$, the overall risk equals 0.0668 regardless of what $P(C1)$ and $P(C2)$ actually are. That's the whole point of Minimax: a guaranteed worst-case error rate that doesn't depend on getting the prior right.

Graphical view

Picture two curves: $R1(\theta)$ starts high at $\theta=4$ and decreases as θ increases; $R2(\theta)$ starts low and increases as θ increases. They cross at $\theta = 5.5$, forming a "V" shape when you look at $\max[R1(\theta), R2(\theta)]$ — the crossing point is exactly the minimum of that maximum, i.e. the Minimax point.

Summary

- **Minimum Error Rate Rule:** pick the class with the highest posterior probability $P(C_i|x)$. This minimizes the average error when priors are known and reliable.
- **Error rate formula:** error = $1 - \max[P(C1|x), P(C2|x)]$ for a single point, or the region-integral formula for the whole space.
- **Minimax Rule:** used when priors are unknown/unreliable. Choose the decision boundary so that the two types of error (miss rate and false alarm rate) are equal — this guarantees the smallest possible worst-case error, independent of the true prior.